Kubernetes Cluster Node Rejoin Guide

This reference document details the procedure to safely remove a worker node from a Kubernetes cluster and rejoin it explicitly utilizing the CRI-O container runtime interface socket.

Step 1: Drain and Delete the Node (Run on Control Plane)

1. Cordon the node to prevent new pod scheduling:

```
kubectl cordon <node-name>
```

2. Drain the node to evict running workloads securely:

```
kubectl drain <node-name> --ignore-daemonsets --delete-local-data --force
```

3. Delete the node object from the cluster API:

```
kubectl delete node <node-name>
```

Step 2: Reset the Worker Node (Run on Target Node)

1. Reset the local kubeadm state to wipe existing configurations:

```
sudo kubeadm reset
```

2. Clean up CNI network and structural environment folders:

```
sudo rm -rf /etc/cni/net.d $HOME/.kube
```

3. Restart and verify CRI-O daemon readiness:

```
sudo systemctl daemon-reload
sudo systemctl restart crio
sudo systemctl enable crio
```

Step 3: Rejoin the Node Using the CRI-O Socket

1. Generate a new join command token on the control plane node:

```
kubeadm token create --print-join-command
```

2. Execute the join command on the worker node, providing the structural path to the CRI-O socket socket endpoint directly:

```
sudo kubeadm join <control-plane-ip>:<port> --token <token> \
--discovery-token-ca-cert-hash sha256:<hash> \
--cri-socket unix:///var/run/crio/crio.sock
```

3. Verify node operational status from the control plane:

```
kubectl get nodes
```